Ansh Kapoor
Chemical Engineering
Indian Institute of Technology Bombay

22B2476
B.Tech.
Gender: Male
DOB: 04/03/2004

Examination	**University**	**Institute**	**Year**	**CPI / %** **Credits**
Graduation	IIT Bombay	IIT Bombay	2026	
Intermediate	CBSE	Sanskar Public School	2022	
Matriculation	CBSE	No. 1 Air Force School	2020	

Pursuing **Minor** Degree in **Computer Science and Engineering** | **Leetcode Knight 2000+ rating**

Scholastic Achievements

- Awarded **Change of Branch** to **B.Tech. Chemical Engineering** for outstanding academic performance (2023)
- Secured **All India Rank 3773** in **Joint Entrance Examination Advanced** among 0.15 million candidates (2022)
- Maintained consistent academic performance with SPI of **9**, **9.57**, **9.64**, and **9.60** over the last **4 semesters** (2025)
- Ranked **1st** in a coding competition during a Bootcamp contest by WNCC and GeeksforGeeks at IIT Bombay (2025)
- Secured **State Rank 10** and **Olympiad Rank 537** in the International Reasoning and Aptitude Olympiad (2018)
- Recipient of Top 1/8 of Passed Students GRADE in CBSE for Mathematics, Physics, Chemistry in 12th Boards (2022)

Professional Experience

Qube Research and Technologies | Quantitative Developer Intern *(Summer 2025)*

- Migrated legacy **GIC** extractor to DataForge with event-based triggering, removing transformed data dependency
- Developed scalable **ENT scripts** for **50** tables to automate ingestion with high data integrity and accuracy
- Created a reusable library of common functions to streamline GIC datasource processing and **reduce duplication**
- Built a reconciliation script to verify **100%** consistency between legacy & new extractors using final DB checks
- Implemented a generic **PyChecker check** for daily index count validation across dev and test environments
- **Fixed multiple cases of historical data corruption** during migration, ensuring clean production deployment
- Updated importer **regex** patterns to fetch files from legacy and new DataForge extractors for seamless migration

Key Projects

Pore Scale Diffusion | Lattice Boltzmann Method *(Summer 2024)*

Nail Your Summers | Guide: Prof. Amol Subhedar

- **Developed** a 2D Unsteady diffusion solver in **C++** to simulate pore-scale diffusion phenomena within **lithium-ion batteries**, encompassing diverse geometries including cylindrical, trapezoidal, rectangular and irregular
- Achieved **95% accuracy** by comparing code results with analytical solutions obtained from differential equation
- Visualized concentration profiles and flux using **Python**, analyzing results for different boundary conditions
- Utilised **ImageJ** to convert **real Battery intersection** into obstacle matrices, facilitating simulation setup
- Presented **Report** and **contributed to research advancements** in porous media modeling and simulation

Uber Backend API | *Self-Project* *(2024)*

- Designed a high-performance ride service API in **Node.js** + **TypeScript** with **MongoDB** for data storage
- Modeled **users**, **drivers**, and **rides** collections with **geospatial queries** in MongoDB for low-latency matching
- Built **WebSocket**-based real-time location & status update system for bi-directional client–server communication
- **Dockerized** services, prepared deployment builds for **AWS EC2** & added unit tests with **Jest** and load testing

Airline Booking System | *Microservices* | *Self-Project* *(2025)*

- Developed microservices backend for **Flights**, **Bookings**, & **Auth** using Node.js, Express, Sequelize with PostgreSQL
- Designed **normalized schemas** and implemented **cross-service associations** to support booking workflows
- Built **idempotent booking APIs** with **concurrency control** and **transactions** to prevent double booking
- Integrated **Kafka** for async notifications and **real-time availability updates**, ensuring eventualconsistency
- **Dockerized** services, configured **Nginx** as reverse **API gateway** & performed **k6** load testing for scalability

LeetCode Clone | *Self-Project* *(2024)*

- Built a secure code-judge backend with **Fastify**, sending submissions to **BullMQ** (**Redis**) for async processing
- Architected sandboxed execution using programmatic **Docker** container creation per submission (**Java**/**C++**)
- Ensured **secure execution isolation**, strict **timeouts**, and safe **resource usage limits** for all submitted code
- Implemented worker orchestration, **result persistence**, and retry/backoff for scaling and reliable async verdicts

Data Structure and Algorithms *(May '23 - Jul '23)*

Summer of Science | Institute Technical Council, IIT Bombay

- Proficient in a wide range of **algorithms** and **data structures** commonly used in competitive programming.
- Skilled in optimizing solutions for **time & space complexity** through adept algorithm and data structure selection.
- Solved **950+** questions on LeetCode and ranked among the **Top 175** coders of **IIT Bombay** at GeekforGeeks.

Predictive Maintenance System for Chemical Factory Equipment *(Autumn 2023)*

Course Project DS203 | Guide: Prof. Vinay Kulkarni | Centre for Machine Intelligence and Data Science

- Developed a machine learning model with **95% accuracy** to predict vibration levels in a chemical factory plant
- Analyzed a dataset of **200+ parameters** and **10,000+ records**, addressing multicollinearity with VIF analysis
- Conducted feature selection using stepwise **multiple linear regression**, identifying **12 significant parameters**
- Developed a real-time alert system integrated with the ML model to automatically notify about vibration issues

Digital Kinetics Monitoring *(Spring 2023)*

Course Project CL208 | Guide: Prof. Sanjay Mahajani | IIT Bombay

- Engineered a real-time digital kinetics monitor using **Arduino** & **ESP32** to capture & transmit pH sensor data.
- Integrated **WiFi module** & ThingSpeak **API** for remote monitoring & data visualization of reaction progress
- Analyzed pH sensor data to calculate **reaction order** and **rate constant** assuming elementary kinetics
- Implemented sensor **calibration** and data **validation** protocols to ensure accuracy of experimental results
- Conducted **10+ experiments** under different reaction conditions to determine the reaction kinetics

Line follower Robot *(Autumn 2022)*

Course Project MS101 | Guide: Prof. Joseph John | Department of Electrical Engineering, IIT Bombay

- Engineered a **Line Following Robot** by leveraging **Fusion 360** for chassis design, incorporated **3-D printing** and **laser cutting** techniques, and utilized **AutoCAD** to streamline and enhance the **design** facilitation process
- Developed an **Arduino code**, calibrated **IR sensors**, and employed **motor drivers** to ensure optimal performance
- Installed a high-efficiency **vacuum cleaner** on top of the chassis with a **suction fan** and a sealed dirt container

Data Quality Improvement for Solar Power Generation Site *(Autumn 2023)*

Course Project DS203 | Guide: Prof. Vinay Kulkarni | Centre for Machine Intelligence and Data Science

- Processed current data with **80000+** data points using moving averages & **winsorization** and performed **EDA**
- Performed feature selection using heatmaps, **Variance Inflation factor** and **p-value** to select key parameters
- Trained and applied **Random Forest Regression** model with cross-validation and **hyperparameter tuning**

Positions of Responsibility

Mentor | Seasons of Code | Web and Coding Club IIT Bombay *(2024)*

- Mentored over **10** students in algorithm design, problem-solving techniques, significantly enhancing their skills
- **Created and managed a Website** providing comprehensive resources for algorithm learning, coding practice
- Created practice problems, and study materials enhancing student proficiency in data structures and algorithms
- conducted **regular coding challenges** and workshops on advanced topics like dynamic programming & graph

Winter in Data Science Mentor | Analytics Club IIT Bombay *(2024)*

- Mentored 8 students on **Data Preparation ML Models**, fostering collaboration and skill development
- Taught advanced **EDA techniques** and strategic model building with a focus on **minimal variables**
- Prioritized hands-on **feature selection** and problem-solving, ensuring a practical ML application approach
- Organized regular discussions, provided constructive feedback, ensuring a supportive learning environment

Courses Undertaken

Computer Science & Data Science	Computer Programming and Utilization, Programming for Data Science, AI and Data Science, Introduction To Machine Learning, Speech and Natural Language Processing and the Web, Computer Networks, Introduction to Cryptography
Mathematics	Calculus, Linear Algebra, Differential Equations, Introduction to Numerical Analysis
Chemical	Introduction to Transport Phenomena , Chemical Engineering Thermodynamics , Process Fluid Mechanics, Material and Energy Balances, Chemical Reaction Engineering, Separation Processes, Fundamentals of Heat and Mass Transfer

Technical Skills

Languages	C++, python, JavaScript, TypeScript, C, SQL, Spark, LaTeX, Arduino
Backend	Node.js, Express.js, Fastify, REST APIs, Microservices, System Design (HLD/LLD)
Databases	PostgreSQL/MySQL, MongoDB(NoSQL), Redis, Sequelize(ORM), Mongoose(ODM)
DevOps	Docker, Docker Compose, Kubernetes, CI/CD, Nginx, AWS, Load Balancing
Tools	Git, WebSockets, gRPC, Message Queues (Kafka, BullMQ), JWT Authentication

Extracurricular Activities

- Completed a year-long training as **Volunteer** at **NSS** Green Campus IIT BOMBAY
- Appeared for **Techkriti Open School Championship** IIT Kanpur 2019
- Participated in Central Board of Secondary Education **(CBSE) Regional Science Exhibition** 2019
- Successfully qualified **Technical Summer School** workshop under Learners' Space IITB in **Data Structure and Algorithms**, **Exploring RTGs & Solar Panels in Rovers**, **Transition to Electric**
- Participated in Defence Research Development Establishment **(DRDE) Science Exhibition** 2018
- Awarded 2nd position in **Sanskrit Antrakshri** inter House Competition in 2018
- Engineered **Python Web crawler** for recursive internal link exploration in **Winter in Data Science** 2024
- Been a part of **Techfest** coordination, which is the largest Science and Technology festival of Asia
- Organiser — **Mood Indigo** (2022): Managed judge transport, crowd control, & timekeeping for Jam Event